Air Quality Visualization And Forecasting Of Air Quality Index Using A Hybrid Machine Learning Model (Random Forest, XGBoost, LSTM With ARIMA Residual Modelling)

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INTRODUCTION

- Rapid urbanization and industrialization have significantly increased air pollution levels, making accurate prediction of the **Air Quality Index (AQI)** essential for public health management, environmental monitoring, and sustainable urban planning. Pollutants such as PM2.5, PM10, NO_2, SO_2, CO, and O_3 exhibit complex, nonlinear, and time-dependent behavior, which makes AQI forecasting a challenging task.
- Traditional statistical and machine learning approaches rely on historical data to estimate AQI levels; however, these methods often fail to capture nonlinear relationships and temporal dependencies present in air quality data, limiting their effectiveness during extreme pollution events and seasonal variations.
- To address these challenges, this study proposes a **hybrid machine learning framework** integrating **Random Forest, XGBoost, and Long Short-Term Memory (LSTM)** models to improve AQI prediction accuracy. The ensemble approach reduces individual model bias and variance, while **ARIMA-based residual modeling** is employed to capture remaining temporal dependencies in prediction errors.
- Furthermore, **SHAP (SHapley Additive exPlanations)** is utilized to enhance model interpretability by quantifying the contribution of each pollutant to AQI predictions, enabling better understanding and decision-making.

"The Need For Decentralized And Sustainable Air Quality In Smart Cities"

- Introduction rapid urbanization industrialization
  - Emission causes localized air quality variation [1], city authorities to airborne pollutants is strongly linked to respiratory diseases [2] because of this city require fine spatial ,temporal data resolution [3].
- Limitations of traditional centralized model for machine learning
  - Security and Privacy: Relying on centralized data aggregation is resource-intensive and inherently compromises data privacy [4], [5], [6], [7], [8]
  - Lack of Spatial Awareness: Traditional methods underestimate the varied toxicity of pollutants across regions and fail during extreme pollution events . Their lack of spatial distribution awareness limits their utility in metropolitan areas [4], [5], [6], [7].
  - Poor Sustainability: Sustainable urban environments rely on optimal energy and power consumption metrics [9]. Traditional models [4], [5], [6], [7] are environmentally harmful due to their high energy demands and assumption of centralized infrastructure, which makes them poorly suited for the fragmented data flows of modern cities.

Problem Formulation

- The problem arises in Predicting AQI due to nonlinear relationships and temporal dependencies in the data . Traditional models like ARIMA capture linear patterns, while machine learning models such as Random Forest and XGBoost model nonlinear relationships but do not inherently capture time-dependent patterns. LSTM networks handle temporal dependencies but may leave residual errors can limit the overall performance of the forecasting system.
- Therefore, this study proposes a hybrid modeling approach that combines **Random Forest, XGBoost, and LSTM** to capture both nonlinear and temporal patterns in the data. The residual errors from the combined model are further modelled using **ARIMA** to enhance prediction accuracy.
- Evaluate model performance using statistical metrics such as **RMSE,MAE,**$\boldsymbol{R^2}$
- Visualization AQI's and Residuals via plotting **Error** created by forecasting as residuals with **ARIMA-residual.**
- In addition, **Shapley Additive explanations (SHAP)** are employed to interpret the contribution of each pollutant to the AQI predictions.

Methodology

- The study begins with **data preprocessing**, where missing values are handled using linear interpolation and pollutant values are converted into AQI using CPCB breakpoints.
- Next, **XGBoost is used to predict future pollutant concentrations**, capturing nonlinear relationships in the data.
- Using these predicted pollutants, **AQI is estimated through three models: Random Forest, XGBoost, and LSTM**, where each model captures different patterns such as nonlinear behavior and temporal dependencies.
  The outputs of these models are then combined using **ensemble averaging** to reduce bias and improve prediction accuracy.
- Further, **residual errors from the ensemble model are also modeled using ARIMA** to capture remaining temporal patterns and refine predictions.
  Finally, **SHAP analysis is applied to interpret the contribution of each pollutant**, making the model transparent and reliable.

DATA PREPROCESSING

Data processing is a crucial step in development of any ML and Time series forecasting model. Real-world air quality datasets often contain inconsistencies such as missing values, varying scales among features, and temporal misalignment. To address these issues as it can cause drastic affect on the accuracy and reliability of predictive model.

In this study the three primary preprocessing techniques are:

❑ **Handling Missing Data:**

Air quality monitoring systems frequently experience data gaps due to sensor malfunctions, maintenance activities, or transmission errors. Missing values can lead to biased model training and reduced predictive performance.

In this study, **linear interpolation** is used to estimate missing values. The method assumes that the change between two data points are linear and then fill the missing values accordingly.

❑ **Mathematical Representation**

If x_t is a missing value between two known observations x_{t_1} and x_{t_2}, the interpolated value is calculated as:

$$\boldsymbol{x_t = x_{t_1} + \frac{(x_{t_2} - x_{t_1})}{(t_2 - t_1)} \times (t - t_1)}$$

❑ **Advantage:**

- Preserves temporal continuity.
- Simple and computationally efficient.
- Avoids introduction of artificial noise.

❑ Feature Selection :

Selection relevant features helps improve model performance, reduce computational complexity, and enhance interpretability. For AQI , the relevant pollutant concentration are the primary determinant of air quality index.

SELECTED FEATURES

The following pollutants were selected as input variables based on their significance in AQI computation:

- PM2.5 ($\mu g/m^3$)
- PM10 ($\mu g/m^3$)
- NO_2 ($\mu g/m^3$)
- SO_2 ($\mu g/m^3$)
- CO (mg/m^3)
- O_3 ($\mu g/m^3$)

The target variable for prediction is the selected features which will give **FINAL AQI**

Note: the feature selection as data is temporal the chances of getting a strong correlation is very less.

❑ Advantage:

- Reduces model complexity.
- Enhances prediction accuracy.

❑Feature Scaling using Min-Max Normalization:

The pollutant variable in the dataset are measured in different units and have different ranges (e.g., PM2.5 in µg/m³ and CO in mg/m³). Models such as LSTM are sensitive to different scaling of input features, as unscaled data can lead to slow convergence and cause numerical instability.

Min-Max Normalisation

Min-Max scaling transform the original data into a fixed range by using activation functions like tanh or sigmoid, which operate within specific ranges (e.g., -1 to 1 or 0 to 1) that also without distorting the feature relation.

❑ Mathematical Representation

$$\boldsymbol{x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}}$$

where:

- X is the original value,
- $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of the feature.

❑Advantage:

- Enhances convergence of neural networks.
- Ensures equal contribution of all features.
- Improves numerical stability.

MODEL DEVELOPMENT AND PERFORMANCE

❑ ADF Test For Stationarity:

The Augmented Dickey Fuller (ADF) test is a formal statistical test use to check the stationarity of any time series dataset.

- **Stationary Series:** The statistical properties (mean, variance) do not change over time. These are easier to predict.
- **Non-Stationary Series:** These exhibit trends or seasonal patterns, making them unpredictable unless transformed .

❑Working principles:

The test evaluates the null hypothesis H_0 that the data is non stationary:

- **If the p-value < 0.05, you reject the null hypothesis; the data is stationary.**
- **If the p-value > 0.05, you fail to reject the null; the data is non-stationary**

❑DEFINE

- **ADF Statistics:** This is the "Test Statistic." It measures how strongly the data depends on a trend. The more negative this number is, the more likely the data is stationary (stable).
- **p-value**: This represents the probability that your data has non-stationary points. In statistics, we typically look for a value below 0.05 to claim success.
- **Critical Levels (1%, 5%, 10%):** these are "benchmarks" or hurdles. To prove your data is stationary, your ADF Statistic must be smaller (more negative) than these values.
  - **1%: High certainty.**
  - **5%: Standard scientific certainty.**
  - **10%: Low certainty.**

Table1: ADF test [DL030]

ADF Statistics	-17.647
p-value	3.734545571876759e-30
1% critical level	-3.43
5% critical level	-2.86
10% critical level	-2.566

Table 2: ADF test [DL024]

ADF Statistics	-19.293
p-value	0.0
1% critical level	-3.431
5% critical level	-2.861
10% critical level	-2.566

Table 3: ADF test [BR005]

ADF Statistics	-12.167
p-value	1.4435857035609038 e-22
1% critical level	-3.43
5% critical level	-2.86
10% critical level	-2.566

Table[1] ADF statistic is -17.64 is very less than all critical levels , and it is most negative that make result highly significant. The p-value is 3.73 *10^-30 which is small and converges near to 0 which means the dataset is STATIONARY.

Table[2] ADF statistic is -19.293 is very less than all critical levels , and it is most negative that make result highly significant. The p-value is which is 0 which means the dataset is STATIONARY.

Table[3] ADF statistics is -12.167 is is very less than all critical levels , and it is most negative that make result highly significant. The p-value 1.44 *10^-22 which is small and converges near to 0 which means the dataset is STATIONARY.

Evaluation Metric for Pollutants:
Evaluation metrics are quantitative measures used to assess the performance, accuracy, and reliability of machine learning models. They are essential for comparing models.

❑ Mean Squared Error(MSE):

It is used to quantify the average squared difference between the actual and predicted AQI values.

- **Mathematical Formula**

$$\textbf{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

❑ Interpretation:

MSE Value	Interpretation
Close to 0	Excellent predictive accuracy
Moderate	Acceptable performance
High	Poor model performance

❑ Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) is a widely used statistical metric for evaluating the accuracy of machine learning models. It measures the square root of the average of the squared differences between the actual observed values and the predicted values

❑ Mathematical Formula

$$\textbf{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Where:

- n**= Number of observations**
- y_i**= Actual (observed) value**
- $\hat{y}_i$**= Predicted value**
- $(y_i - \hat{y}_i)$**= Prediction error (residual)**

Relationship with MSE

- RMSE is directly derived from the Mean Squared Error (MSE):
- $\text{RMSE} = \sqrt{\text{MSE}}$

❑ Coefficient of Determination (R^2)

The Coefficient of Determination (R^2) measures the proportion of variance in the observed AQI values that is explained by the predictive model. It provides an indication of the model's goodness-of-fit, with values closer to 1 signifying superior explanatory power.

❑ Mathematical Formula

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

❑ Interpretation

R^2 Value	Interpretation
1	Perfect fit
0.7 – 0.9	Strong predictive capability
0.5 – 0.7	Moderate fit
< 0.5	Weak model performance
Negative	Model performs worse than the mean

Table 1:Pollutants Performance Metric For BR005:

XGBoost Pollutant Prediction Metrics	MSE	RMSE	R^2
PM 2.5 (ug/m3)	14.03	3.75	0.89
PM 10 (ug/m3)	26.12	5.11	0.87
NO2 (ug/m3)	8.04	2.84	0.97
SO2 (ug/m3)	2.22	1.49	0.79
CO (mg/m3)	0.22	0.34	0.84
OZONE (ug/m3)	7.84	2.80	0.84

Table 2: Pollutants Performance Metric For DL030:

XGBoost Pollutant Prediction Metrics	MSE	RMSE	R^2
PM 2.5 (ug/m3)	27.08	41.67	0.82
PM 10 (ug/m3)	66.63	94.28	0.73
NO2 (ug/m3)	8.84	13.58	0.74
SO2 (ug/m3)	2.84	5.01	0.78
CO (mg/m3)	0.41	0.66	0.71
OZONE (ug/m3)	10.50	15.81	0.81

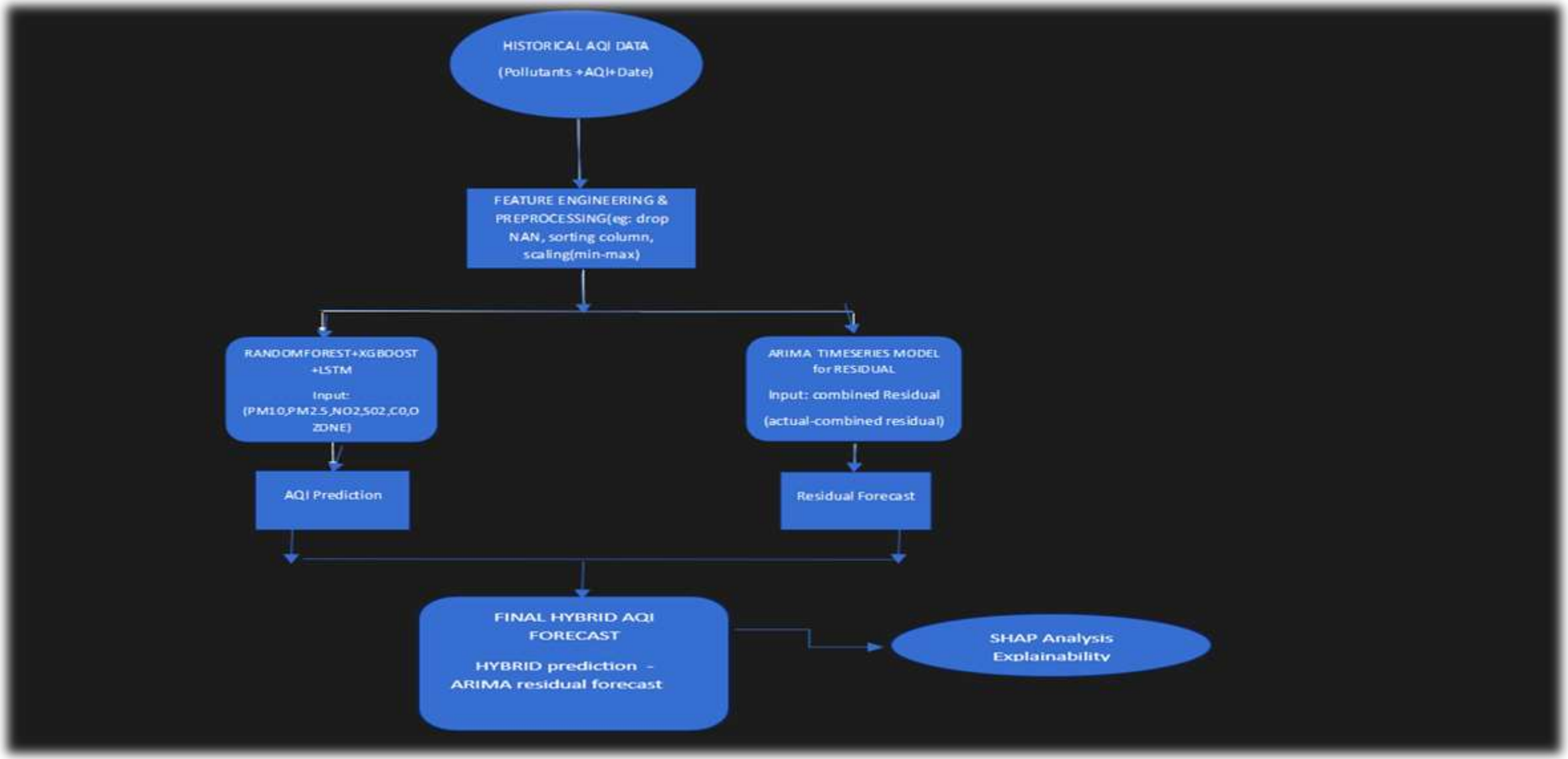
Table 3: Pollutants Performance Metric DL024

XGBoost Pollutant Prediction Metrics	MSE	RMSE	R^2
PM 2.5 (ug/m3)	31.20	5.59	0.83
PM 10 (ug/m3)	60.73	7.79	0.77
NO2 (ug/m3)	0.00	0.00	1.00
SO2 (ug/m3)	2.45	1.57	0.75
CO (mg/m3)	0.41	0.64	0.70
OZONE (ug/m3)	9.15	3.02	0.60

Pollutant-Wise Interpretation:

- Table[3] : **High accuracy overall:** XGBoost shows strong performance for most pollutants ($R^2 \approx 0.70–0.83$), indicating good predictive capability. **Best prediction:** NO_2 achieves near-perfect fit ($R^2 = 1.00$), suggesting extremely accurate modeling (possible overfitting or low variance). **Relatively weaker areas:** Ozone and CO have lower R^2 values (0.60–0.70), indicating higher variability and reduced prediction accuracy.
- Table[2] : **Consistent performance:** XGBoost achieves strong prediction accuracy across pollutants ($R^2 \approx 0.71–0.82$), indicating reliable modeling. **Best predictions:** PM2.5 and Ozone show higher accuracy ($R^2 \approx 0.81–0.82$), capturing trends effectively. **Moderate variation:** PM10 and NO_2,CO have slightly lower R^2 (~0.73–0.74), suggesting room for improvement in capturing variability.
- Table[1] : **Strong overall accuracy:** XGBoost performs well for most pollutants ($R^2 = 0.70–0.83$), indicating reliable predictions. **Perfect fit for NO_2:** $R^2 = 1.00$ with zero error suggests extremely high accuracy (possible overfitting or low variability).L**ower performance for CO:** $R^2 = 0.79$ indicates higher uncertainty and scope for model improvement.

FLOWCHART


HISTORICAL AQI DATA
(Pollutants +AQI+Date)
FEATURE ENGINEERING & PREPROCESSING(eg: drop NAN, sorting column, scaling(min-max)
RANDOMFOREST+XGBOOST +LSTM
Input: (PM10,PM2.5,NO2,SO2,CO,O ZONE)
ARIMA TIMESERIES MODEL for RESIDUAL
Input: combined Residual
(actual-combined residual)
AQI Prediction
Residual Forecast
FINAL HYBRID AQI FORECAST
HYBRID prediction - ARIMA residual forecast
SHAP Analysis Explainability

Comparison (Random Forest , XGBoost, LSTM) models

I. Random Forest Regressor:

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions. It effectively captures nonlinear relationships between pollutants and AQI.

- **Mathematical Representation:**

$$\hat{Y}_t^{RF} = \frac{1}{B}\sum_{b=1}^{B} T_b\,(X_t)$$

- **where T_b represents the bth decision tree.**

II. XGBoost (XGB)

XGBoost is a gradient boosting algorithm that builds trees sequentially, minimizing prediction errors using a regularized objective function. It offers high predictive accuracy and computational efficiency.

- **Objective Function:**

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

III. Long Short-Term Memory (LSTM)

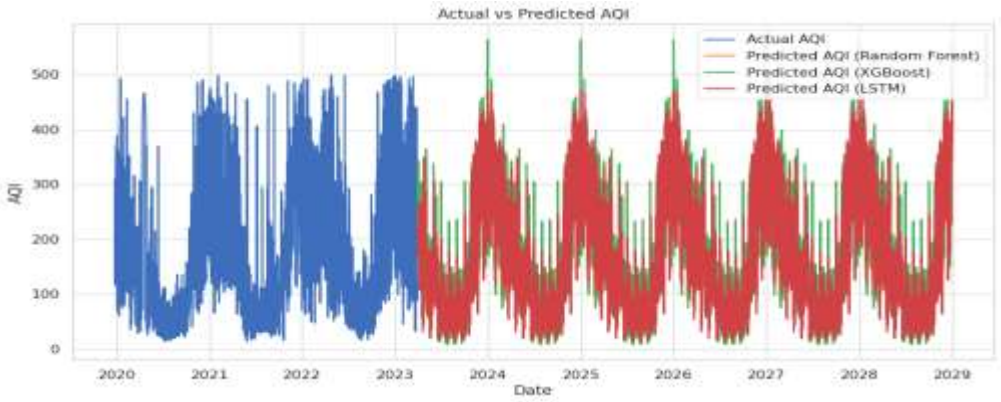
LSTM networks are a type of recurrent neural network designed to capture long-term dependencies in sequential data. The gating mechanisms—input, forget, and output gates—enable the model to retain relevant temporal information.

- **Prediction Function:**

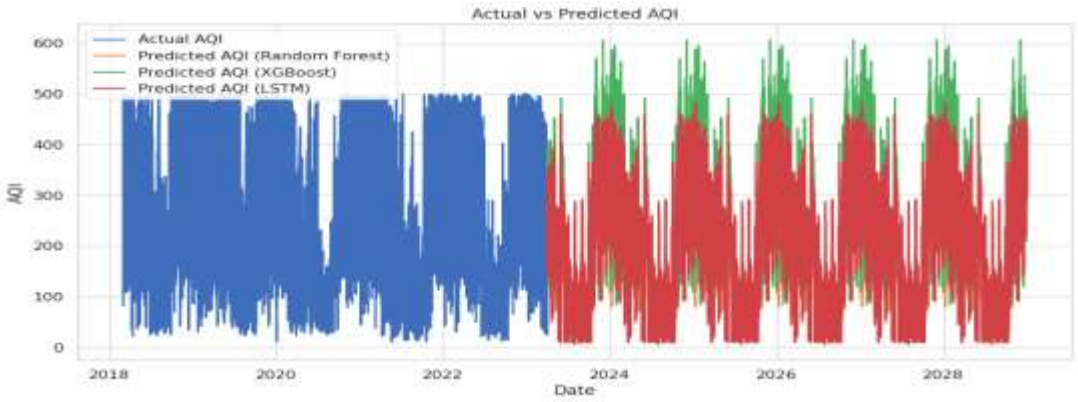
$$\hat{Y}_t^{LSTM} = f_{LSTM}(X_{t-n}, \dots, X_t)$$

COMPARISON GRAPHS ACTUAL AQI TO PREDICTED (RFR,XGBOOST ,LSTM)

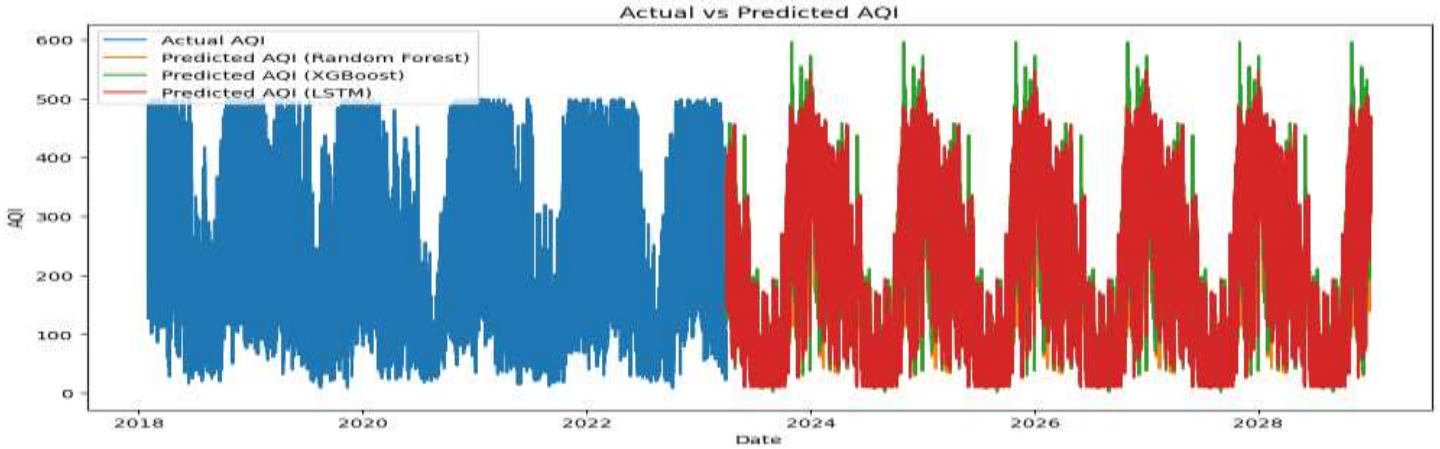
GRAPH 1: FOR BR005



GRAPH 2: FOR DL030



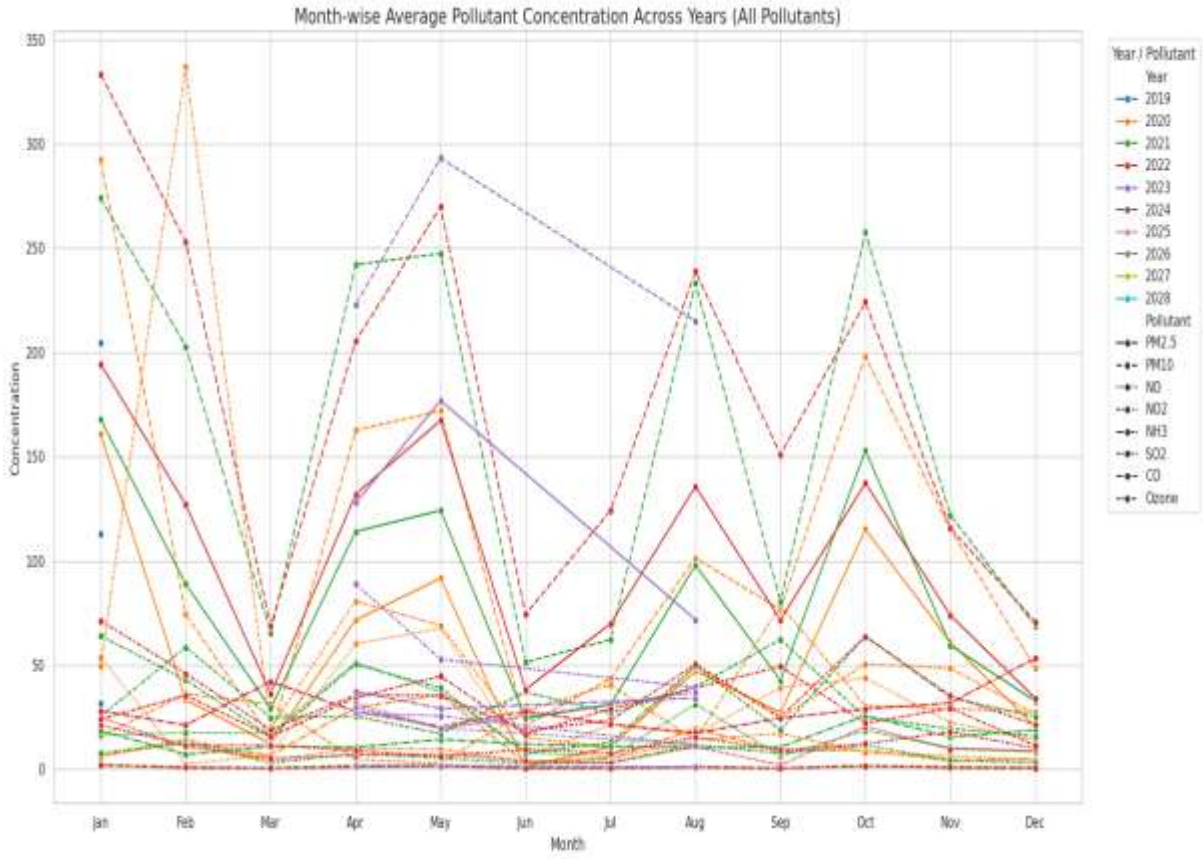
GRAPH 3 : FOR DL024



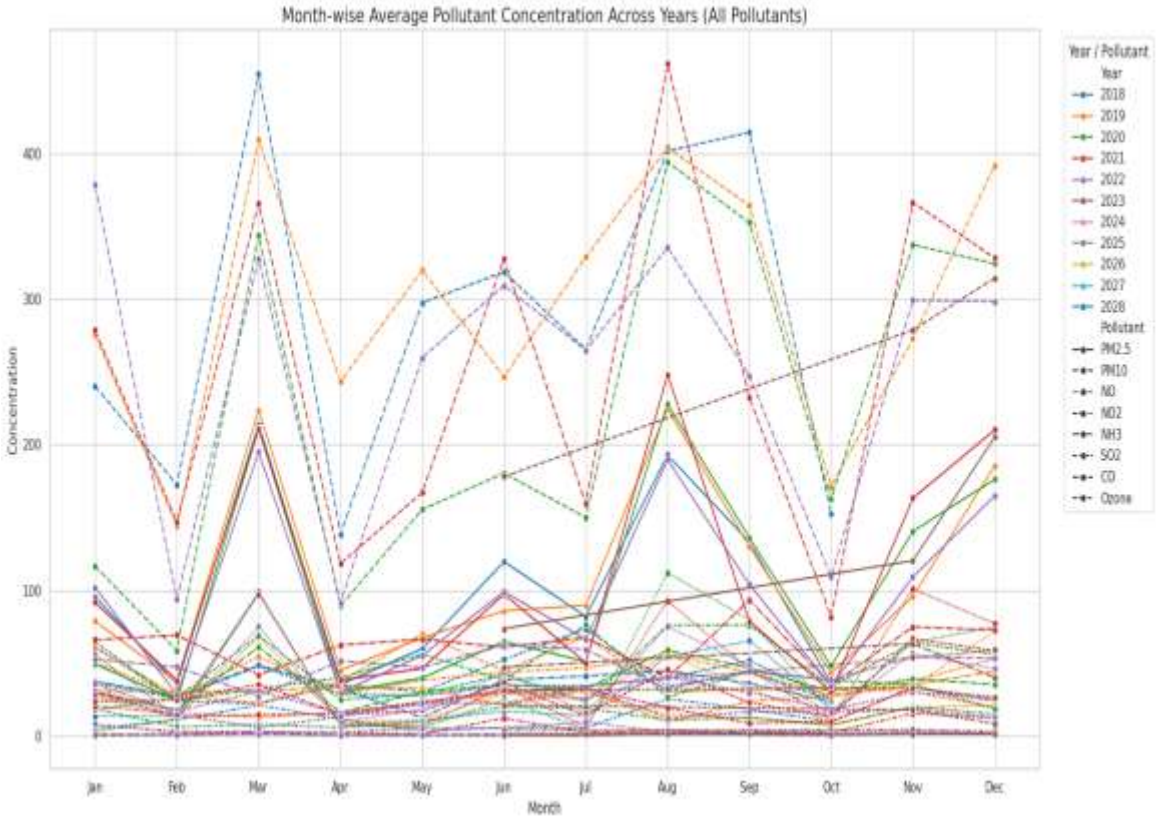
The predictions from the three models closely align with each other, indicating consistency and robustness in forecasting, with XGBoost and LSTM capturing sharper peaks while Random Forest appears slightly smoother. Predicted values span from low to very high AQI levels (up to ~600), suggesting potential future pollution events.

Pollution concentration (original dataset vs predicted dataset)

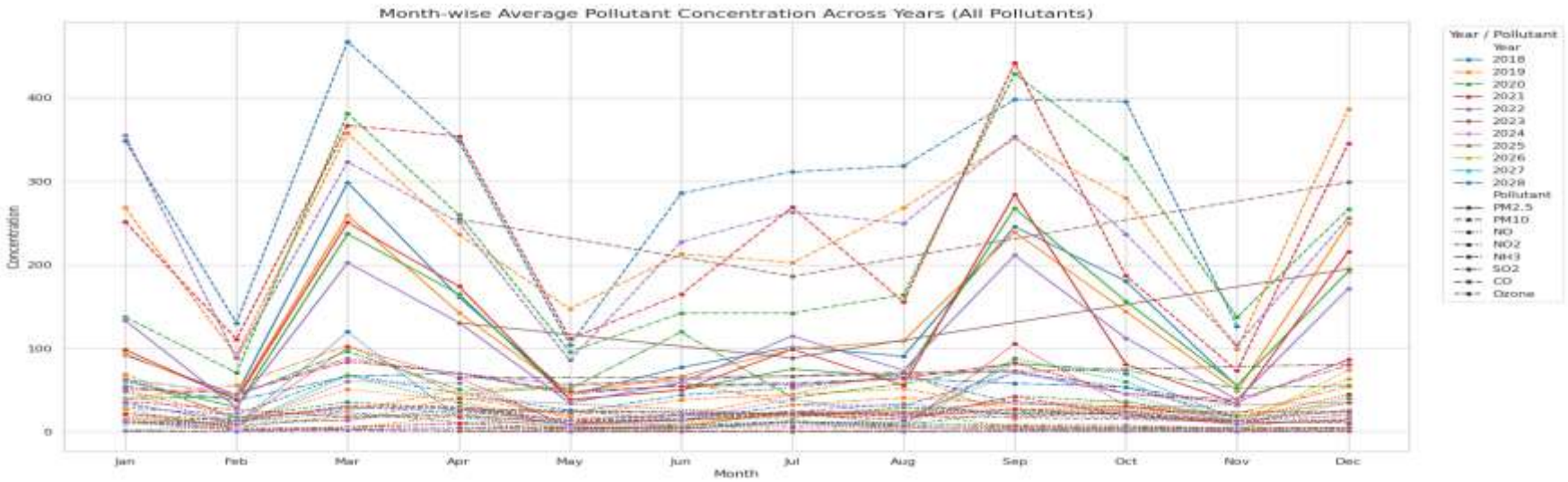
Graph 1: Comparison Of Pollutants [br005]



Graph 2: Comparison Of Pollutants[DL030]



Graph 3: comparison of pollutants [DL024]



Graph 1: pm 2.5 and pm10 are in highest concentration compared to other pollutants. In winters [oct to dec] and in prime summers [may –June] and decline in monsoon.
Graph 2: PM2.5 and PM10 consistently show the highest concentration levels ,making them major contributor with respect to other pollutants .Pollution spikes are prominent in monsoons [aug-sept] and winter [nov-dec] ,while it dips around February and October.
Graph 3:PM2.5 and PM10 consistently show the highest concentration levels ,making them major contributor with respect to other pollutants .Pollution spikes are prominent in monsoons [aug-sept] and winter [nov-dec] ,while it dips around February and October.

ENSEMBLE MODEL PERFORMANCE

Fig I: COMPARISON GRAPH FOR PREDICTION MODEL [BR005]

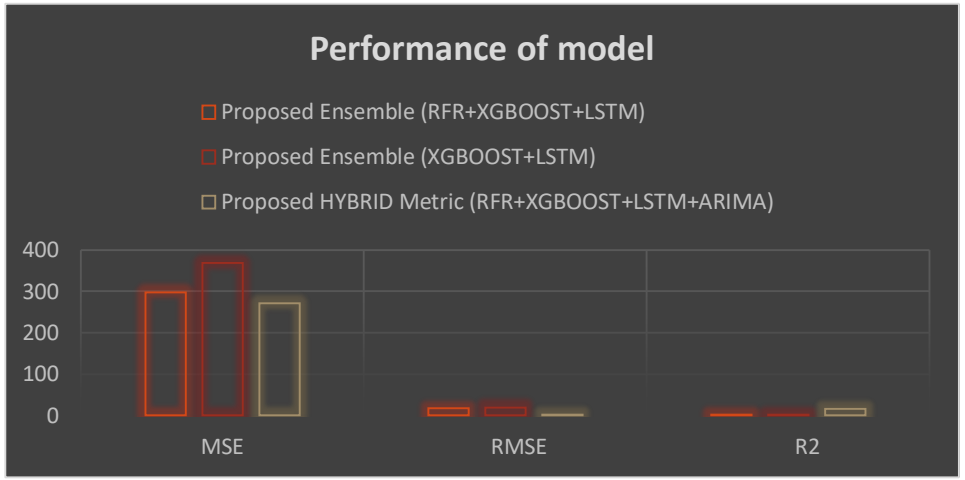
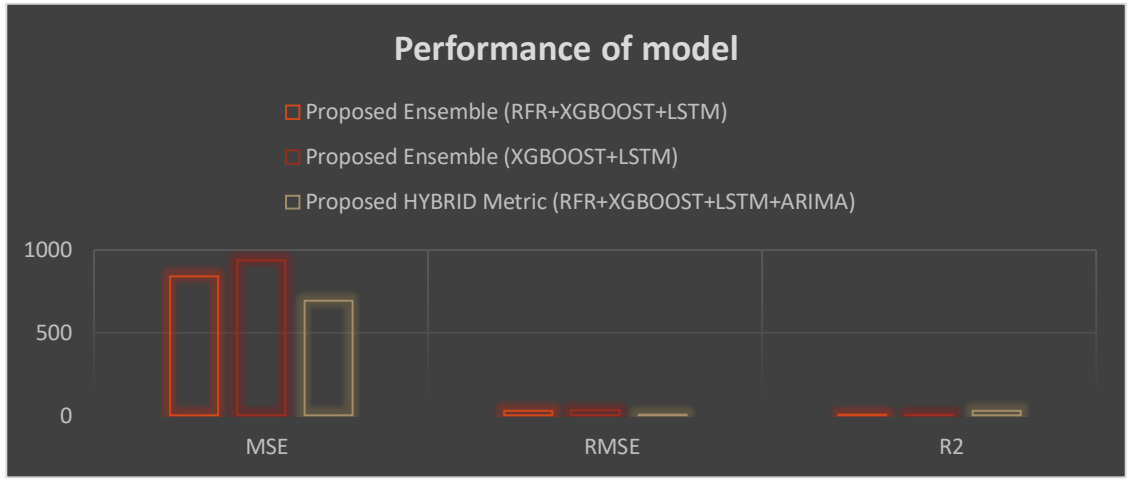


Fig II: COMPARISON GRAPH FOR PREDICTION MODEL [DL024]



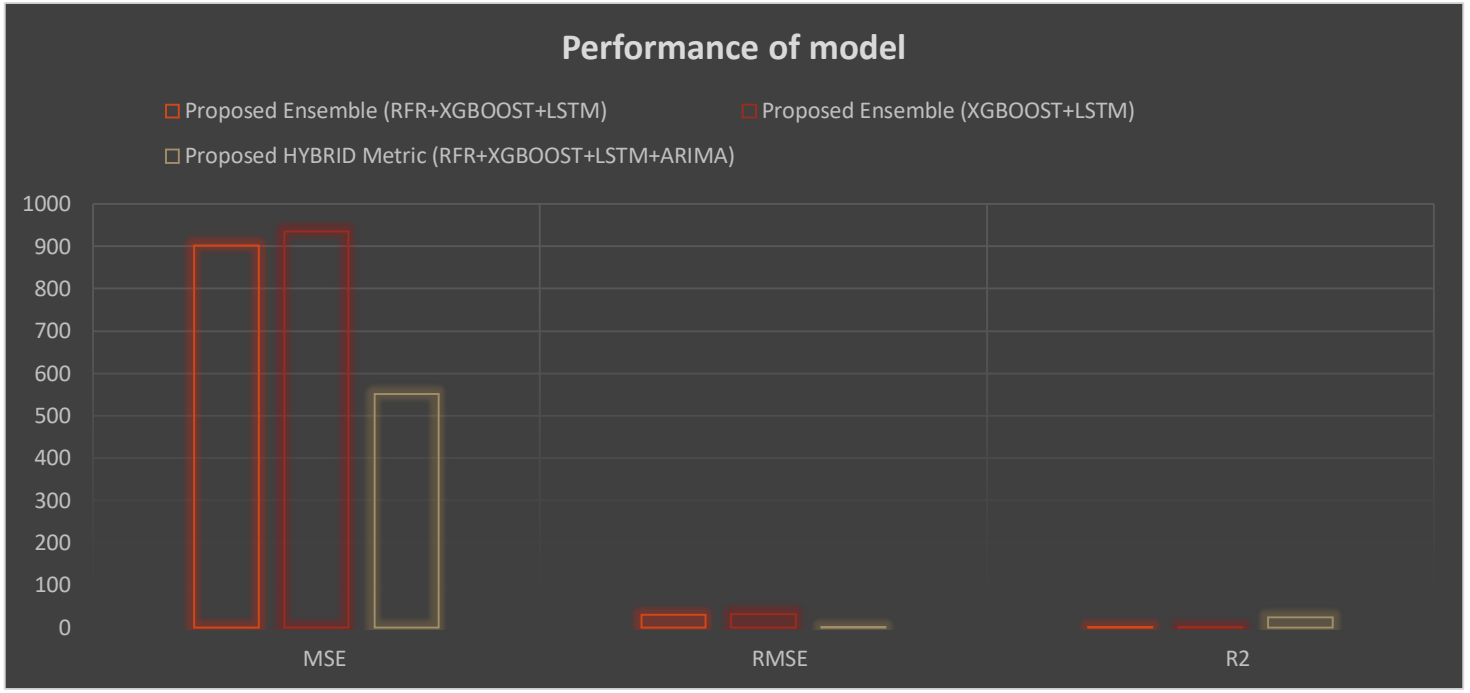
INTERPRETATION:

Fig [I]: Combining all three models significantly improves predictive accuracy.it is the **best-performing model** among the three. The **lowest MSE and RMSE** indicate superior prediction accuracy. The inclusion of **ARIMA-based residual modeling** captures remaining linear temporal patterns in the errors. Although the R^2 remains 0.98 (similar to the first ensemble), the reduction in error metrics confirms the effectiveness of the hybrid approach.

Fig [II]: This is the best-performing model among the three. By incorporating ARIMA-based residual modeling, the hybrid approach captures remaining linear temporal patterns in the ensemble's errors, resulting in:

- The lowest MSE and RMSE, indicating superior accuracy.
- The highest R^2, reflecting enhanced explanatory capability

Fig III: EVALUATION METRIC FOR DL030



INTERPRETATION :

- The hybrid is the best-performing model among the three. By incorporating ARIMA-based residual modeling, the hybrid framework captures remaining linear temporal patterns in the ensemble's prediction errors. This results in:
- A substantial reduction in MSE and RMSE, indicating improved accuracy.
- An increase in R^2, demonstrating better explanatory capability.

RESIDUAL ANALYSIS

❑Residual Calculation

The residuals represent the unexplained component of the ensemble prediction

$$\boldsymbol{e_t = Y_t - \hat{Y}_t^{ENS}}$$

❑ ARIMA Model

The **Auto-Regressive Integrated Moving Average (ARIMA)** model, denoted as ARIMA(p, d, q), is used to model residual patterns. The optimal parameters are selected using **Auto-ARIMA** based on information criteria such as AIC.

❑Final Hybrid Prediction

The final AQI forecast is obtained by adding the predicted residuals to the ensemble prediction:

$$\hat{Y}_t^{FINAL} = \hat{Y}_t^{ENS} + \hat{e}_t$$

$$\hat{Y}_t^{ENS} = AQI_{RFR} + AQI_{XGBOOST} + AQI_{LSTM}$$

RESULTS: From hybrid model like *rfr+ xgboost+ lstm+ ARIMA model* approach integrating this model via training and testing ,even with high-performing model ,some pattern in the residual (the differences between actual and predicted AQI) remained Unexplained. So the respective figures explains those errors

RESIDUAL PLOT FOR TESTED COMBINED(ACTUAL) RESIDUAL VS ARIMA - Residual over time

Fig 1: BR005

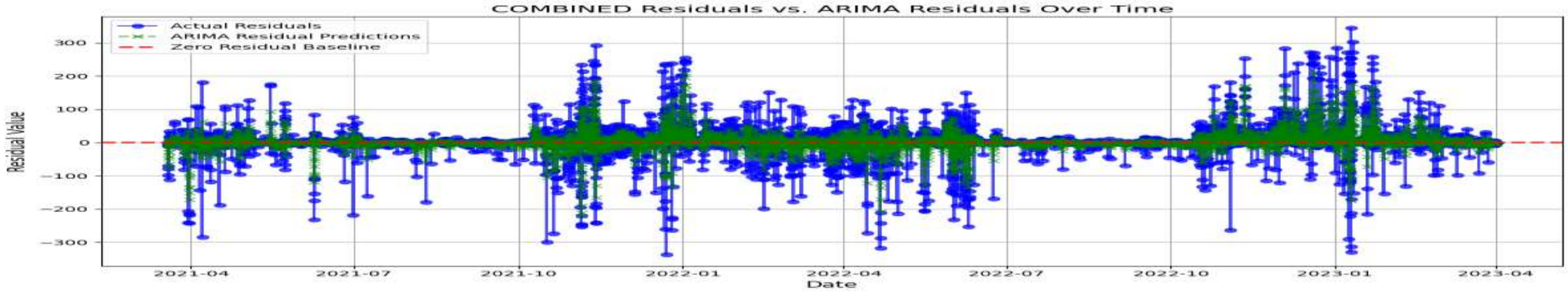


Fig 2:DL024

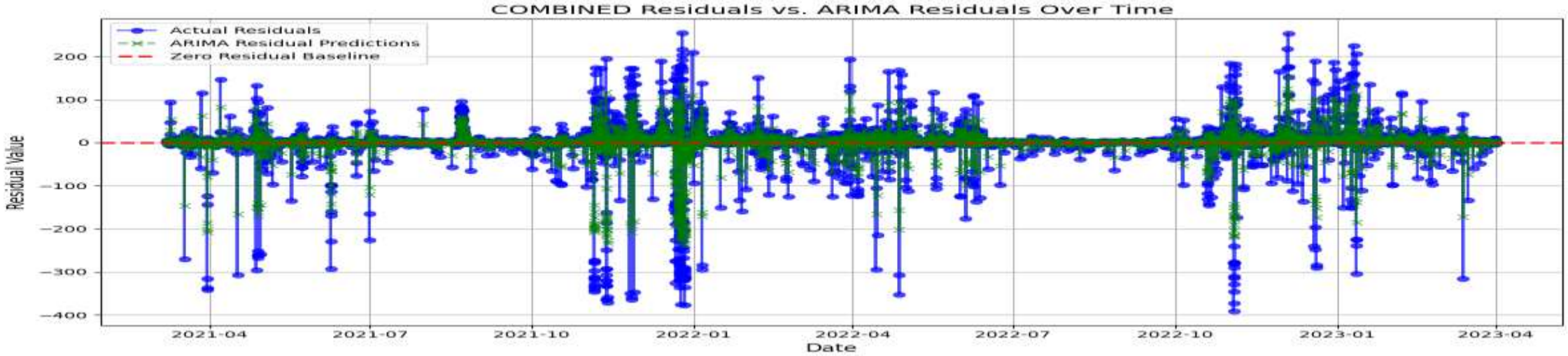
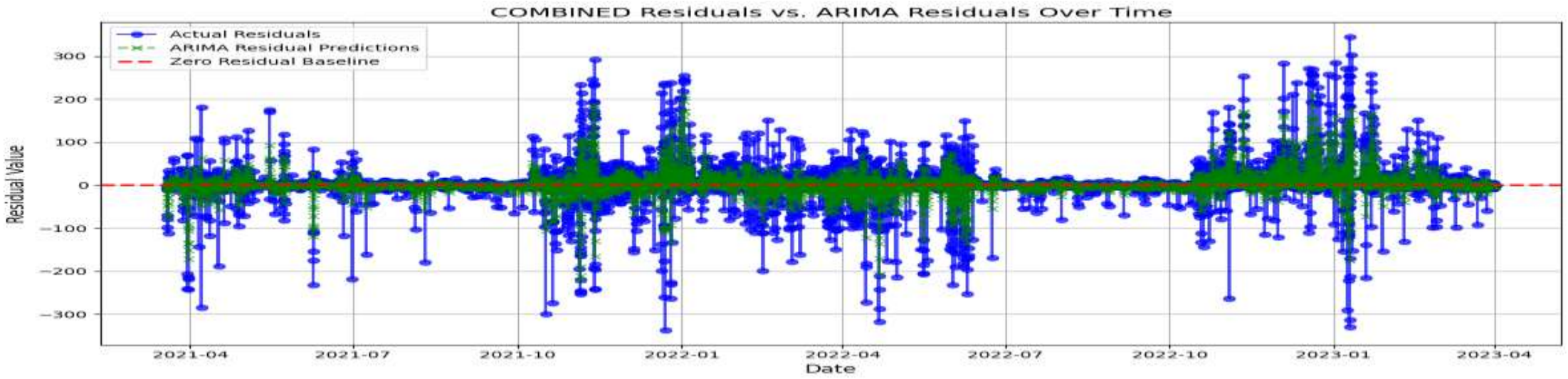


Fig 3: DL030



INTERPRETATION:

Fig[1],Fig[2],Fig[3] :The actual residuals fluctuate around the zero baseline, indicating that the ensemble model is largely unbiased but occasionally experiences large errors during extreme pollution events . The ARIMA-predicted residuals closely follow the pattern of the actual residuals ,it capture the remaining temporal dependencies in prediction. ARIMA-based residual modeling enhances the accuracy and reliability of the final hybrid AQI forecasts.

SHAP ANALYSIS FOR MODEL PERFORMANCE:

- SHAP SUMMARY

the hybrid model +ARIMA model is subject to some limitation. To start with hybrid model relies greatly on the presence and quality of historical pollutant concentration data; incomplete or inconsistent measurements may decrease prediction accuracy. However, the model designed is extensible. Exogenous variables such as meteorological parameters (e.g., temperature, wind speed, humidity), population density, and policy interventions can significantly influence air quality dynamics.

1.SHAP PERFORMANCE OF BR005

index	Rank	Pollutant	Most Impactful Feature ▲	Feature Score (Mean Abs SHAP)	Reason
3	4	Ozone (ug/m3)	hour	12.988945960998535	Hourly patterns (e.g., traffic, industrial activity) are the most significant factor.
0	1	PM10 (ug/m3)	month	73.98988342285156	Monthly variations (seasonal patterns) are the most significant factor.
1	2	PM2.5 (ug/m3)	month	42.78194808959961	Monthly variations (seasonal patterns) are the most significant factor.
2	3	NO2 (ug/m3)	month	23.807598114013672	Monthly variations (seasonal patterns) are the most significant factor.
4	5	SO2 (ug/m3)	month	2.929384708404541	Monthly variations (seasonal patterns) are the most significant factor.
5	6	CO (mg/m3)	month	0.441470742225647	Monthly variations (seasonal patterns) are the most significant factor.

2. SHAP PERFORMANCE OF DL030

1 to 6 of 6 entrie

Rank	Pollutant	Most Impactful Feature	Feature Score (Mean Abs SHAP)	Reason
1	PM10 (ug/m3)	month	83.0460205078125	Monthly variations (seasonal patterns) are the most significant factor.
2	PM2.5 (ug/m3)	month	56.52363586425781	Monthly variations (seasonal patterns) are the most significant factor.
3	Ozone (ug/m3)	hour	20.31132698059082	Hourly patterns (e.g., traffic, industrial activity) are the most significant factor.
4	NO2 (ug/m3)	month	10.239503860473633	Monthly variations (seasonal patterns) are the most significant factor.
5	SO2 (ug/m3)	year	5.427920341491699	Yearly trends (long-term changes) are the most significant factor.
6	CO (mg/m3)	hour	0.4534723460674286	Hourly patterns (e.g., traffic, industrial activity) are the most significant factor.

3. SHAP PERFORMANCE OF DL024

Rank	Pollutant	Most Impactful Feature	Feature Score (Mean Abs SHAP)	Reason
1	PM10 (ug/m3)	month	85.15240478515625	Monthly variations (seasonal patterns) are the most significant factor.
2	PM2.5 (ug/m3)	month	66.07247161865234	Monthly variations (seasonal patterns) are the most significant factor.
3	Ozone (ug/m3)	hour	14.912816047668457	Hourly patterns (e.g., traffic, industrial activity) are the most significant factor.
4	NO2 (ug/m3)	month	8.823256492614746	Monthly variations (seasonal patterns) are the most significant factor.
5	SO2 (ug/m3)	year	3.362262487411499	Yearly trends (long-term changes) are the most significant factor.
6	CO (mg/m3)	month	0.4128373861312866	Monthly variations (seasonal patterns) are the most significant factor.

CONCLUSION

- This study transforms complex air pollution data into meaningful insights, revealing clear seasonal patterns and key pollutant impacts.
- The hybrid model (XGBoost + RF + LSTM + ARIMA) improves prediction accuracy by capturing both nonlinear and time-based behavior.
- Comparison of actual data with predicted data by XGBoost, RFR,LSTM . We can see DL030 has more sharpness which shows increase in pollution in future trends with seasonality and temporal patterns with compare to DL024 AND BR005.
- ARIMA –residual prediction is better than residuals it capture errors which remained undiagnosed in normal residual it give more precision less error.
- Overall, the findings provide a reliable, data-driven framework to support better environmental monitoring and policy decisions.

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